Dumps Outline

Grade 03 - 04

1. **Introduction of AI**
   - Simple Definition of AI
   - AI in Our Daily Lives
2. **Human vs. AI**
   - Comparing Human and Machine Decision-Making
3. **Why is AI Cool?**
   - Describe the AI work
4. **How Does AI Learn?**
   - Explained the process of AI Learning
5. **AI Around Us**
   - AI technologies: Search Engine, Social Media, Self Driving Cars
6. **Advantages of AI**
   - Benefits of AI
7. **History of AI**
   - Timeline of AI

Grade 05 - 06

1. **What is AI**
   - Explanation of AI and Everyday Examples
2. **Types of AI**
   - Narrow AI
   - General AI
3. **How AI Makes Decisions**
   - Learning from Data and Examples
4. **AI in Various Industries**
   - Education, Agriculture, Healthcare
5. **Creativity with AI**
   - Practical Activity: AI Tools in Daily Life
6. **Basic Ethics of AI**
   - What is AI Ethics

Grade 07 - 08

1. **What is Machine Learning?**
   - Basic Concepts Explained
2. **Basic Concepts**
   - Data
   - Algorithms
   - Model
3. **Types of Machine Learning**
   - Supervised and Unsupervised Learning
4. **How Machine Learning Works?**
   - LifeCycle of Machine Learning
5. **ML Real-World Examples**
   - Machine Learning in Voice Assistants and Search Engines
6. **Solving Problems with ML**
   - Apply ML Concepts to Real-World Scenarios

Grade 09 - 10

1. **Introduction to AI**

   - Definitions of AI and ML
   - Its purpose
   - How it mimics human cognitive functions
   - Discussion on Weak AI (narrow AI), Strong AI (general AI), and Superintelligence, with examples and implications.

2. **Basics of Algorithms in AI**

   - Introduction to common algorithms used in AI, including their purpose in decision-making and problem-solving.

3. **Applications of AI**

   - Real-world examples of AI in fields like healthcare (diagnostics), finance (fraud detection), entertainment (recommendation systems), and transportation (traffic management).

4. **Machine Learning Basics**

   - Introduction to machine learning, explaining how systems improve their performance on tasks through experience.

5. **Types of Machine Learning**

   - Overview of:
     - Supervised Learning: Learning from labeled data.
     - Unsupervised Learning: Identifying patterns in unlabeled data.
     - Reinforcement Learning: Learning through trial and error with rewards and penalties.

6. **Machine Learning Models**
    - Overview of key models including regression and classification, and their applications in predictive analytics.
7. **Data and Its Role in AI**
    - The significance of data for training AI models, including data collection, preprocessing, and quality importance.
8. **Neural Networks Basics**
    - Introduction to artificial neural networks, including their structure (neurons and layers) and their use in modeling complex patterns.
9. **Natural Language Processing (NLP)**
    - Basics of how machines understand, interpret, and generate human language, with applications like translation and sentiment analysis.
10. **Robotics and AI**
    - Exploration of how AI enhances robotics for tasks like automation, navigation, and interaction with humans.
11. **Ethical Issues in AI**
    - Discussion of the ethical considerations surrounding AI technologies, such as privacy concerns, algorithmic bias, and job displacement.
12. **Chatbots and Virtual Assistants**
    - How AI powers conversational agents, enabling them to assist users with queries and tasks through natural language understanding.

13. **Autonomous Vehicles**

- Basics of self-driving technology, including the role of AI in navigation, obstacle detection, and decision-making.

14. **Facial Recognition Technology**

- Overview of how facial recognition works, its applications in security and personalization, and associated privacy issues.

15. **Pattern Recognition**

- Understanding how AI identifies and classifies patterns in data, applicable in image and speech recognition.

16. **History and Evolution of AI**

- Brief timeline of AI development, highlighting key milestones and figures that shaped the field.

Grade 11 - 12

1. **Basics of Deep Learning**
   - Introduction to deep learning concepts, architecture of deep neural networks, and comparison with traditional machine learning.
2. **Artificial Neural Networks (ANNs)**
   - Structure and functioning of ANNs, activation functions, and their role in pattern recognition.
3. **Convolutional Neural Networks (CNNs)**
   - Specialized neural networks for image processing, including layers, pooling, and real-world applications.
4. **Recurrent Neural Networks (RNNs)**
   - Understanding RNNs for sequence data, applications in language processing, and the concept of long short-term memory (LSTM).
5. **Natural Language Processing (NLP) Techniques**
   - Key techniques such as tokenization, stemming, lemmatization, and sentiment analysis.
6. **Computer Vision Fundamentals**
   - Overview of how AI enables machines to interpret and understand visual data from the world, including applications in image recognition and video analysis.

7. **Data Science and Big Data in AI**

   - Role of data science in AI, handling large datasets, and tools used for big data analytics.

8. **Reinforcement Learning Concepts**

   - Basics of reinforcement learning, agents, environments, rewards, and applications in game playing and robotics.

9. **Feature Engineering and Data Preprocessing**

   - Techniques for selecting and transforming features to improve model performance, including normalization and encoding.

10. **AI in Cybersecurity**

    - Application of AI techniques in threat detection, malware analysis, and cybersecurity strategies.

11. **AI Algorithms**

    - Overview of algorithms such as Decision Trees, K-Nearest Neighbors, and Support Vector Machines (SVM), with their applications.

12. **Ethics and Bias in AI**

    - Discussion on ethical concerns, biases in AI models, and the importance of fairness and transparency.

13. **AI in Predictive Analytics**

    - Use of AI to predict outcomes based on historical data, applications in business and finance.

14. **Genetic Algorithms**

- Basics of genetic algorithms, their principles, and applications in optimization problems.

15. **AI in Cloud Computing**

- How cloud services enhance AI capabilities, including computing power and data storage.

16. **Introduction to Quantum Computing in AI**

- Overview of quantum computing and its potential impact on AI algorithms and processing capabilities.

17. **Autonomous Systems and Drones**

- Use of AI in autonomous systems, including navigation, control, and applications in various industries.